「Def.
A group $D_{2n} \overset{def}{=} \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$ is the dihedral group of order $2n$.

「Prop. Let D_{2n} be the dihedral of order $2n$.
1). For any $x \in D_{2n} \setminus \langle r \rangle$, $rx = xr^{-1}$.
2). For any $x \in D_{2n} \setminus \langle r \rangle$, $|x| = 2$.
3). $D_{2n} = \langle s, sr \rangle$.
4). If $n = 2k$ is even with $n \geq 4$, then r^k has order 2 and $\langle r^k \rangle = \{1, r^k\} = Z(D_{2n})$.
5). If n is odd with $n \geq 3$, then $Z(G) = \{1\}$.
Proof. Let $x \in D_{2n}$ be given. Then $x = r^a s$ for some $a = 0, 1, \cdots, n-1$, by $x \notin \langle r \rangle$.
So, $rx = r^{a+1} s = r^a \cdot r \cdot s = r^a s r^{-1} = x \cdot r^{-1}$.
And, $x^2 = r^a s r^a s = r^{a-1} s r^{-1} r^a s = \cdots = s r^{-a} \cdot r^a \cdot s = s^2 = 1$, and $x \neq 1$,
so $|x| = 2$. ⌋ The 3) is given by $\langle s, sr \rangle \subseteq D_{2n}$ and $s\,sr = r \in \langle s, sr \rangle$, so
$\langle s, sr \rangle \supseteq \langle s, r \rangle = D_{2n}$. ⌋
To determine the Center of D_{2n}, observe that; given $a \in \{0, 1, \cdots, n-1\}$,
$$r^a \in Z(D_{2n}) \iff r^a s = s r^a = s r^{-a} \iff r^{2a} = 1,$$ by the cancelation law.

Since $r^{2a} = 1$ if and only if $n \mid 2a$, so $r^a \in Z(D_{2n})$ if and only if
$$a = \begin{cases} 0 & \text{if } n \text{ is odd} \\ 0, \frac{n}{2} & \text{if } n \text{ is even.} \end{cases}$$
And, given $x \in D_{2n} \setminus \langle r \rangle$, $x \in Z(D_{2n})$ implies that $xr = rx = xr^{-1} \Rightarrow r^2 = 1$.
But, it contradicts to $|r| = n \geq 3$. Consequently, for $n \geq 3$,
$$Z(G) = \begin{cases} \{1\} & \text{if } n \text{ is odd} \\ \{1, r^k\} & \text{if } n = 2k \text{ is even} \end{cases}$$
☑

$\ulcorner$ Prop. If n is even, then $D_{2n} \cong \langle (1\ 2\ \cdots\ n), (2\ n)(3\ n-1)\cdots(\frac{n}{2}\ \frac{n}{2}+2) \rangle \leq S_n$
If n is odd, then $D_{2n} \cong \langle (1\ 2\ \cdots\ n), (2\ n)(3\ n-1)\cdots(\frac{n+1}{2}\ \frac{n+1}{2}+1) \rangle \leq S_n$.

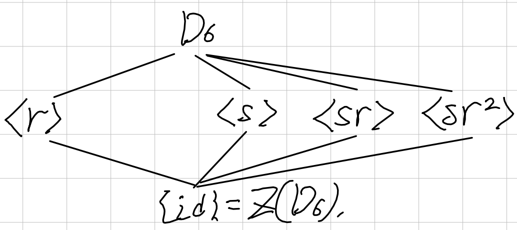
Proof In case of n is even number, observe that

$$(1\ 2\ \cdots\ n)(2\ n)(3\ n-1)\cdots(\tfrac{n}{2}\ \tfrac{n}{2}+2)(1\ 2\ \cdots\ n)$$
$$=(1)(2\ n)(3\ n-1)\cdots(\tfrac{n}{2}\ \tfrac{n}{2}+2)$$

Moreover, $(1\ 2\ \cdots\ n)^n = id$ is clear, and since $(2\ n)\cdots(\frac{n}{2}\ \frac{n}{2}+2)$ is product of disjoints two-cycles, so it has order 2.
So, corresponding to $r=(1\ 2\ \cdots\ n)$ and $s=(2\ n)\cdots(\frac{n}{2}\ \frac{n}{2}+2)$,
it satisfies the representation of the definition. Similarly, n is odd, can shown. $\boxtimes$

D_6.

By the theorem above, $D_6 \cong S_3$. Now, explicitly,



ΓD_8

ΓD_{10}